A hybrid system for navigation in GPS-challenged environments: case study

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BIOGRAPHY

Professor Chris Rizos is a graduate of the University of New South Wales (UNSW), Sydney, Australia, obtaining a Ph.D. in satellite geodesy. He is currently the head of the School of Surveying & Spatial Information Systems at UNSW. Rizos has been researching the technology and applications of GPS since 1985 and over a decade ago established the Satellite Navigation and Positioning (SNAP) group, today the largest and best known academic GPS and wireless location technology R&D laboratory in Australia. Chris is the vice president of the International Association of Geodesy (IAG), a member of the governing board of the International GNSS Service (IGS), a Fellow of the IAG and a Fellow of the Australian Institute of Navigation.

Dr. Dorota Brzezinska is a Professor and leader of the Satellite Positioning and Inertial Navigation (SPIN) Laboratory at The Ohio State University. She received her M.S. and Ph.D. in Geodetic Science from The Ohio State University. Her research interests cover kinematic positioning with GPS, precision orbit determination of GPS/LEO, INS/GPS/image/LiDAR integration, mobile mapping technology, and robust estimation techniques. Since 2003, she has served at the ION Council in various capacities, and is currently Eastern Region Vice-President. She is Vice-President of the International Association of Geodesy (IAG) Commission 4, Positioning and Applications and chair of IAG Sub-Commission 4.1, Multi-sensor Systems. Dorota is the 2005 recipient of the US Geospatial Intelligence Foundation (USGIF) Academic Research Award and the 2005 ION Thurlow Award. She is a Fellow of the International Association of Geodesy.

Dr. Charles Toth is a Senior Research Scientist at The Ohio State University Center for Mapping. He received an M.S. in Electrical Engineering and a Ph.D. in Electrical Engineering and Geo-Information Sciences from the Technical University of Budapest, Hungary. His research expertise covers broad areas of 2D/3D signal processing, spatial information systems, high-resolution imaging, surface extraction, modelling, integrating and calibrating of multi-sensor systems, multi-sensor geospatial data acquisition systems, and mobile mapping technology. He is chair of ISPRS WG I/2 on LiDAR and InSAR Systems and serves as the Director of the Photogrammetric Application Division of ASPRS.

Dr. Andrew G. Dempster is the Director of Research in the School of Surveying & Spatial Information Systems at UNSW. He led the team that developed Australia's first GPS receiver in the late 1980s and has been involved with satellite navigation ever since. His current research interests are GNSS receiver design, GNSS signal processing, and new location technologies.

Dr. Yong Li is a senior research fellow at the SNAP Lab within the School of Surveying & Spatial Information Systems, UNSW. His current interests include multi-navigation sensor integration, attitude determination, GPS receiver technique, optimal estimation theory and applications, and FPGA technology and its application to navigation.

Nonie Politi is a graduate of the School of Electrical Engineering & Telecommunications at UNSW. He obtained a Bachelor degree in Telecommunication Engineering and a M.Eng.Sc. in Electronics. Nonie is currently working as a research assistant at the School of Surveying and Spatial Information Systems, UNSW, researching the new Locata positioning technology.

Dr. Joel Barnes is Director of Navigation R&D for Locata Corporation, and is also a Senior Visiting Research Fellow at the SNAP Lab, UNSW. Joel has assisted in the development of the Locata receiver and testing of the Locata technology since 2000, whilst working at UNSW as a Research Fellow. He joined the Locata Corporation in 2007 and is responsible for navigation algorithm research and development.

Dr. Hongxing Sun is a post-doctoral researcher in the SPIN Laboratory, at The Ohio State University. He received a Bachelors degree in Geodesy, and M.S. and Ph.D. degrees in Photogrammetry from Wuhan University, China. His research interests include precise kinematic positioning with GNSS, GNSS/INS integration for direct platform orientation, and integrated multi-sensor geospatial data acquisition systems.

ABSTRACT

Reliable and continuous navigation in GPS-challenged environments is only possible with integrated systems that include self-contained sensors, which facilitate independent navigation during GPS outages. Traditionally, an inertial navigation system (INS) has been integrated with GPS to allow bridging during GPS gaps. The recently developed RF-based navigation and positioning system "Locata" offers an attractive augmentation alternative in situations where GPS satellite geometry is poor or the signal availability is limited.

In cooperation with Locata Corporation, a Canberra-based company, School of Surveying and Spatial Information Systems (at the University of New South Wales - UNSW), and the Satellite Positioning and Inertial Navigation (SPIN) Laboratory (of The Ohio State University - OSU), have assembled a working prototype of a hybrid system based on GPS, inertial navigation and Locata receiver technology to provide seamless and reliable navigation to support vehicle guidance and control for mining, construction, mobile and GIS mapping, and industrial applications.

For experimental purposes, a dual IMU system, based on the navigation grade H764G and a tactical grade CMIGITS2, were used. In addition, a Locata receiver and a dual-frequency GNSS receiver, the Leica System 1200, were used on a test vehicle at the Locata's Numerella Test Facility, near Canberra, Australia. This test site features areas of both open sky as well as moderate to dense foliage. The test was repeated by mounting the devices on an autonomous electric car, operated on the UNSW campus. The GPS and Locata data were processed separately (for testing the internal consistency) as well in a hybrid solution, resulting in few centimetre-level accuracy for each coordinate component, depending primarily on GPS availability and the geometry between the test vehicle and the LocataLites, as well as the level of multipath.

INTRODUCTION

The determination of the position and orientation (or "pointing direction") of a device (or platform to which it is attached), to high accuracy, in all outdoor environments, using reliable and cost-effective technologies is something of a "holy grail" quest for navigation researchers and engineers.

However, this research has identified two classes of applications that place stringent demands on the positioning/orientation device: (a) man-portable mapping and imaging systems that operate in a range of difficult urban and rural environments, often used for the detection of underground utility assets (pipelines, cables, conduits), unexploded ordnances and buried objects, and (b) the guidance/control of construction or mining equipment in environments where good "sky view" is often unlikely.

The solution to this positioning/orientation problem is increasingly seen as being based on an integration of several technologies; satellite (GPS) and terrestrial (Locata) ranging systems, inertial navigation systems (INS), laser guidance/scanning systems, and even electro-optical devices such as surveyors' "Total Stations" or "laser scanners". Each has its shortcomings but within an integrated system advantage can be taken of the complementary characteristics of several of these sensor technologies. First feasibility tests on a system that includes many of the abovementioned systems were reported in [1][2], where the authors report on the first triple-integration GPS/INS/Locata results on a vehicle.

Centimeter-level accuracy positioning systems for outdoor use typically have at their core the GPS technology. GPS is in fact the most effective general-purpose navigation tool ever developed because of its ability to address a wide variety of applications; air, sea, land and space navigation, precise timing, geodesy, surveying and mapping, machine guidance/control, military and emergency services operations, hiking and other leasure activities, personal location and location-based services; using different: (a) receiver instrumentation, (b) operational procedures, and (c) data processing techniques [3]. All require signal availability from a minimum of four GPS satellites.

However, one of the limiting factors in using GPS is the need for direct line-of-sight between the satellites and the ground receiver. In particular, the robustness of positioning is compromised when GPS receivers are near or under trees, in urban/suburban areas, or in deep open-cut mines and construction sites, where there is partial sky view obstruction by buildings and walls. The traditional means of overcoming the gaps in navigation coverage due to satellite signal blockages is to use an INS. An INS is also the most convenient means of determining the orientation of the device/platform. The integration of GPS and INS can, in principle, overcome the defects of standalone INS (sensor errors that grow unbounded with time) and GPS (signal availability requirement) systems.

But, navigation accuracy degrades rapidly if there are no GPS measurements to calibrate the IMU sensor errors [3][4].

A new terrestrial, RF-based, distance measurement technology offers promise of continuous signal coverage, even in difficult urban/rural environments. This technology is known as "Locata" [5].

The Locata approach is to deploy a network of ground-based transceivers that cover an area with strong time-synchronized ranging signals. When a Locata receiver uses four or more ranging signals it can compute a high accuracy position entirely independent of GPS or INS [5]. However, a standalone Locata receiver has its own shortcomings: (a) in some situations it may be difficult to achieve good VDOP due to logistical constraints of placing transmitters (to give a variation in elevation angle between the terrestrial transmitters and the receiver whose positions is to be determined), and (b) as with GPS, multiple receivers/antennas are required to derive orientation information.

What is therefore required is several carefully selected navigation sensor technologies, integrated within a single hardware package, the measurements from which are simultaneously processed to provide continuous, reliable and accurate navigation solutions (i.e. both position and orientation information).

In cooperation with Locata Corp., the School of Surveying & Spatial Information Systems (UNSW) and the SPIN Laboratory (The Ohio State University) have assembled a working prototype of a hybrid system based on GPS, inertial navigation and Locata receiver technology to provide seamless and reliable navigation aimed at supporting vehicle guidance and control, open-pit mining, mobile and GIS mapping, and industrial applications.

For experimental purposes, a dual IMU system, based on the navigation grade H764G and a tactical grade CMIGITS2, were used. In addition, a Locata receiver and a dual-frequency GPS receiver, the Leica System 1200, were placed on the car at the Numeralla Test Facility, near Canberra, Australia. This test site features both open sky scenario, allowing for testing the system's performance in a truly challenging environment. The test was repeated by mounting the devices on the autonomous electrical car, driven on the UNSW campus. In both cases the separation between the rover and the terrestrial transmitters was between a few tens of metres to several kilometres. The GPS and Locata data were processed separately (for testing the internal consistency) as well in a hybrid solution, resulting in few centimetre level accuracy per coordinate, depending primarily on GPS availability and the geometry between the rover and LocataLites, as well as the level of multipath fading.

LOCATA TECHNOLOGY

The School of Surveying & Spatial Information Systems at UNSW has been conducting pseudolite research for many years, and has experimented with them in nonsynchronous and synchronized modes for a variety of applications, using both the GPS L1 frequency as well as the 2.4GHz ISM band frequencies [6-12]. Locata Corporation has developed state-of-the-art RF terrestrial positioning technology ("Locata"), which consists of a network ("LocataNet") of time-synchronized pseudolite-like transceivers ("LocataLites"). UNSW has assisted in the development of the technology through experimental testing and benchmarking. In a relatively open outdoor environment the LocataNet can provide real-time stand-alone kinematic positioning (without a base station) at centimetre-level accuracy [8]. Even in an indoor environment where LocataLite signals arrive at a Locata receiver via non-line-of-sight (LOS) paths (penetrating the walls of buildings), the static positioning quality can be at the sub-centimetre level, and also at the sub-metre level for kinematic positioning [9].

Locata has several advanced features that have been developed over a period of about 8 years, through several technology generations, including a time-synchronized positioning network, network propagation to many LocataLites, improved signal penetration, change of transmitting frequency and signal structure, and spatial and frequency diversity.

In Table 1, the key characteristics of the two generations of Locata technology are listed. Using 2.4GHz not only means the frequency licence is free, but also permits transceiver output power of up to 1 Watt, which means greater operating distances (up to 10km). Using dual-frequency signals changes the initial phase bias resolution from known-point initialization to On-The-Fly (OTF), where the initial phase bias is resolved while the receiver is moving. The higher chipping rate (10MHz) results in less pseudorange multipath error, because the delay in a reflected signal will rarely be more than two chips. The 10Hz measurement rate allows relative high velocities of the receiver.

In terrestrial-based RF-based positioning, multipath error is more severe than with GPS, because the terrestrially transmitted signal arrives at the receiver at a very low (typically less than 10 degrees) or even a negative elevation angle, which can result in severe multipath signal fading. In the second generation Locata system spatial and frequency diversity techniques are employed. Spatial and Frequency diversity are two of the three types of diversity principles (the other being polarization) that

are common practices in terrestrial RF communications to mitigate against signal fading. The LocataLite transceiver uses two spatially separated (usually in the vertical) antennas, which transmit two signals at different frequencies. This gives a cluster of four diverse signals transmitted from one LocataLite. With this diversity technology, Locata kinematic positioning in moderately obstructed environments can provide centimetre-level quality with 100% coverage, as well as consistent geometry and high reliability [7]. The Locata's multipath mitigation technology is very important and relevant to this project, because the operational environments are often vegetated or wooded.

Table 1. Specification summary of Locata's first and second generation systems [6].

Components		First generation	Second generation
Signal structure	Frequencies	Single frequency at GPS L1	Dual frequency 2.4 GHz (80Mhz bandwidth)
	PRN code	C/A (1.023MHz chipping rate)	Proprietary (10MHz chipping rate)
	License requirements	Licensing issues & problem for wide area deployment	None required, FCC compliant
LocataLite (transceiver)	Hardware	FPGA & DDS technology	FPGA & DDS technology with a modular design
	Output power	Several microwatts	Maximum of 1 Watt
	Range	600 meters	10km line-of-sight
Locata (receiver)	Hardware	Zarlink/Mitel based GPS receiver chipset	FPGA technology, modular design
	Measurement rate	1Hz	10Hz
	Initial phase bias resolution	Known point initialization (KPI)	On-The-Fly

TRIPLE INTEGRATION OF GPS/INS/LOCATA

As discussed in the preceding sections, there are both advantages and disadvantages to every navigation sensor. GPS and Locata have high positioning accuracy in open or moderately obstructed environment, but they are sensitive to signal blockage such as the case in dense forests, urban canyons, deep mine pits and indoors. In contrast, INS is totally autonomous, i.e., independent of external signal sources, and has high output rate for position, velocity and attitude, but its navigation error grows rapidly with time.

The most common tool to integrate GPS and INS is the Kalman filter, which forms the basis for multi-sensor integration in this research. The basic Kalman filter applies to linear system models, therefore, several derivatives were developed to cope with the non-linear navigation model, such as the Extended Kalman Filter or the Unscented Kalman Filter [13].

The following discussion of the integration of the GPS/INS/Locata sensors is focused on two aspects: 1) the system state selection, and 2) the measurement model or integration model that decides which information to pass to the filter.

The error state vector consists of a 9-dimensional navigation error state sub-vector $\mathbf{x}_{rv\varepsilon}$ (three for the position, three for the velocity and three for the orientation), an accelerometer error state sub-vector $\mathbf{x}_f$, a gyroscope error state sub-vector $\mathbf{x}_\omega$, and a 3-dimensional gravity disturbance state sub-vector $\mathbf{x}_g$. Of course, other sensor error models can be considered for the gyroscope and accelerometer sensors, such as a combination of random constants, 1st order Gaussian-Markov variables, scale factors, etc. In this case, the state space could have a dimension of more than 30. The objective is to adjust the sensor error model later based on experimental results (if needed); however because of the limitations of observability, it is not yet known whether an augmented error state vector would give better results.

When integrating INS hardware with other sensors, the sensors can not share the same physical location, which would be ideal from a theoretical point of view. Knowing the spatial relationship among the sensors is important to ensure the highest possible navigation performance. The displacement vectors or mounting biases are offsets, also referred to as 'lever arms', from the centre of the IMU to the centres of the other sensors. These lever arm parameters may be included in the Kalman filter and thus can be estimated [14]. However, if the lever arms are precisely measured during the assembly of the system, they do not need to be included in the filter as estimable parameters.

For multiple sensor integration in a Kalman filter there are essentially two types of general models: loosely-coupled and tightly-coupled. The loosely-coupled model uses a decentralized filter that has several sub-filters to process the sub-systems independently. In other words, the Kalman filter solutions from the sub-systems are combined in a Kalman filter that provides the integrated

navigation solution. In contrast, the tightly-coupled model uses a single main filter to process the output of all sensors. In GPS/INS integration, tightly-coupled systems have obvious advantages in environments where GPS signals are frequently lost, because they can rely on the other sensor(s) when GPS positioning becomes impossible [15].

In the tightly-coupled model, the raw observations of all sensors will be input to the main filter. For GPS and Locata, the primary observations will be the carrier phase measurements, as code observation cannot provide the required accuracy. High accuracy GPS positioning needs to address the issue of carrier phase ambiguity. The ambiguity can be treated as an unknown in the Kalman filter, but it may take several minutes to resolve the ambiguity using GPS alone. Using certain ambiguity resolution techniques, however, the ambiguity can be resolved outside the main filter in the GPS/INS high-precision (carrier phase) integration filter [15]. Note that if the ambiguity were to be resolved within the filter, this would increase the number of states of the filter. For the GPS component, ionospheric delay should be included for applications that cover a large area. Ionospheric delay can be resolved using network-based differential techniques, but it will affect the ambiguity resolution for single baseline differential positioning if it is not included in the local solution. The filter is designed either to use, or not to use, ionospheric delay, which can ensure flexibility to accommodate network-based and single baseline differential positioning. The state of the integration filter is therefore:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_{rv\varepsilon} & \mathbf{x}_f & \mathbf{x}_\omega & \mathbf{x}_g & \mathbf{x}_I \end{bmatrix}^T \tag{1}$$

where $\mathbf{x}_I$ refers to ionospheric delay, either used or not used, as a state in the filter, depending upon needs. The system model can be written as:

$$\begin{bmatrix} \dot{\mathbf{x}}_{rv\varepsilon} \\ \dot{\mathbf{x}}_f \\ \dot{\mathbf{x}}_\omega \\ \dot{\mathbf{x}}_g \\ \dot{\mathbf{x}}_I \end{bmatrix} = \begin{bmatrix} \mathbf{F}_{11} & \mathbf{F}_{12} & \mathbf{F}_{13} & \mathbf{F}_{14} & \mathbf{0} \\ \mathbf{0} & \mathbf{F}_{22} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{F}_{33} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{F}_{44} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{F}_{55} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{rv\varepsilon} \\ \mathbf{x}_f \\ \mathbf{x}_\omega \\ \mathbf{x}_g \\ \mathbf{x}_I \end{bmatrix} + \begin{bmatrix} \mathbf{v}_{rv\varepsilon} \\ \mathbf{v}_f \\ \mathbf{v}_\omega \\ \mathbf{v}_g \\ \mathbf{v}_I \end{bmatrix} \tag{2}$$

where ionospheric delay can be considered as a 1st order Gaussian-Markov process, $\mathbf{F}_{55} = \eta \mathbf{I}_n$, and n is the number of double-differenced equations.

As mentioned above, the measurement model in the tightly-coupled model is based on the raw observations. For GPS and Locata, the observations will be the carrier phase observations. The linear measurement equation is:

$$\begin{bmatrix} \Phi_{GPS} \\ \tilde{\tilde{\rho}}_{GPS} \\ \Phi_{LOC} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_{Gr} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{B}_I \\ \mathbf{B}_{Gv} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_P & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{x}}_{rv\varepsilon} \\ \dot{\mathbf{x}}_f \\ \dot{\mathbf{x}}_\omega \\ \dot{\mathbf{x}}_g \\ \dot{\mathbf{x}}_I \end{bmatrix} + \mathbf{v} \tag{3}$$

where Φ_{GPS} is the double-differenced carrier phase after ambiguities are resolved; $\tilde{\tilde{\rho}}_{GPS}$ is the double-differenced Doppler observations; Φ_{LOC} is the Locata carrier phase observation after the initial bias resolution with dual-frequency data; $\mathbf{B}_{Gr}$ is the GPS positioning coefficient matrix; $\mathbf{B}_I$ is the double-differenced ionospheric delay coefficient matrix; $\mathbf{B}_{Gv}$ is the GPS velocity determination coefficient matrix; $\mathbf{B}_p$ is the Locata positioning matrix; and $\mathbf{v}$ is the observation noise. The approximate values for the linearization of the GPS and Locata measurement equations are provided by the INS navigation solution.

The GPS carrier phase ambiguity is solved independently outside the Kalman filter with OTF techniques. The GPS differential positioning coefficient $\mathbf{B}_{Gr}$ matrix remains the same regardless whether or not a network-based differential technique is used. For velocity determination, the double-differenced Doppler observation, $\tilde{\tilde{\rho}}_{GPS}$, is used to eliminate the clock error rate as an unknown (because it is difficult to model this in the filter). The initial carrier phase bias of the Locata is also not included in the filter, because it can be resolved instantaneously with dual-frequency data in the Locata second generation system.

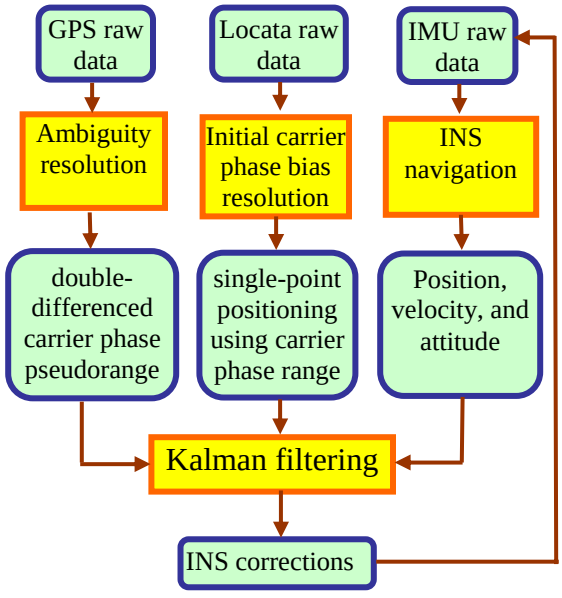


Fig. 1. Workflow of the integrated GPS/ INS/Locata system

The implementation of Eq. (3) in the filter will be flexible, so adjustments can be made to account for actual environmental conditions. The filter is designed with an open interface and is modular in structure, so that components can be added (or removed) from model in Eq. (3). In open sky areas, GPS is sufficient for system positioning, so only its observations need to be processed. In moderately obstructed environments, GPS and Locata observations will be processed. In this case the number of GPS observation equations is limited and sometimes will be less than four. Fig. 1 illustrates the flowchart of triple-integration of GPS, INS and Locata.

MULTI-SENSOR TIME SYNCHRONIZATION

Using the FPGA technology, researchers at the SNAP Lab have developed a time synchronization device for logging GPS and INS data [4,5,16,17]. The device has been modified to log the Locata data and time-align the Locata data with the GPS time frame.

Two time-sync FPGA devices were used in the tests (for logging the GPS, Locata and C-MIGITS data), and all the data was aligned to the GPS time frame for subsequent processing within the Kalman filter. Fig. 2 shows the time-sync FPGA devices used in the tests. Further development will combine all the inputs into one FPGA device.

The time-sync FPGA device is built around the Nios II soft-core on a Stratix EP1s10 device. The GPS pulse-per-second (PPS) signal is required for the time synchronization process and is connected to the device via a BNC socket. The device is currently configured with two time-sync UARTs for INS (or Locata) and GPS input.

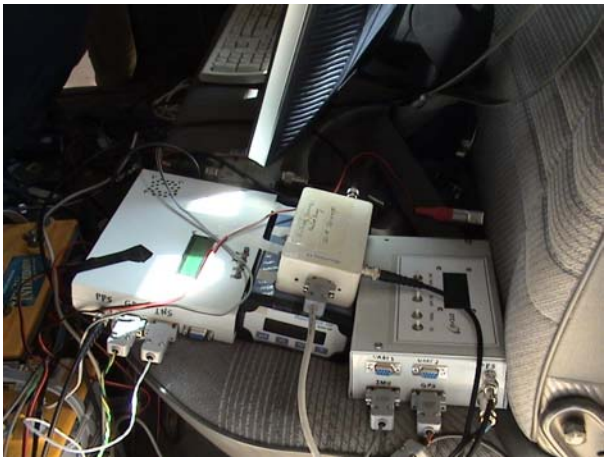


Fig. 2 Time-sync FPGA devices for data logging

The time-sync UART logic is attached to the processor as a memory-mapped peripheral with one interrupt line [17]. The UART must detect transmission, receive the data in serial format, strip off the start- and stop-bits, and store the data word in a parallel format, as well as access a free-running counter that is latched at the start bit of a serial transmission. This count is appended to the incoming byte and placed in a first-in-first-out (FIFO) buffer. The PPS signal along with GPS time data are used in an interpolation algorithm to calculate the time-of-arrival of serial data from the INS or Locata. In this way the INS/Locata data is time-tagged with GPS time.

More details about the time-sync FPGA device can be found in the literature [4].

FIELD TESTS

Test 1 – NTF

The first integration test was conducted at Locata's Numerella Test Facility (NTF), which is located in NSW, Australia, on March 17, 2008. The NTF covers an area of approximately three hundred acres (2.5km x 0.6km) and is ideally suited to 'real-world' system testing over a wide area. At the NTF a number of LocataNet configurations are possible through the installation of permanent antenna towers. The network configuration used for this experiment is illustrated in Fig. 3.



Fig. 3. NTF: LocataLite network

Prior to the test, a special mounting platform was designed and built. The platform, shown in Fig. 4, consists of a two-level metal frame. The bottom level can accomodate two inertial measurement units, while the top level can hold up to four antennas, attached using 5/8" Whitworth bolts. The platform can be easily attached to either the roof of the NTF test vehicle or to the body of UNSW's small electric car (described later).



Fig. 4. Devices setup for the NTF test

The devices that were used in the test include, two Leica dual-frequency GPS receivers (one used as the rover receiver, and the other as the base station), one H764G IMU and one Locata rover unit. The GPS antenna and the Locata antenna were mounted with the IMU together on the top of a truck. The GPS data rates were set to 1Hz. The average length of GPS differential baseline is about 1.2km. The GPS observation conditions were good during the testing period. The Locata data rate was set to 10Hz, while IMU data rate was 256Hz, and both of them were synchronized with the GPS time as described earlier.

The GPS/INS data were first processed in tightly-coupled mode first. The trajectory is depicted in Fig. 5. The standard deviation of position, velocity and attitude are shown in Figs. 6-8 respectively. The position error state in the Kalman Filter is shown in Fig. 9.

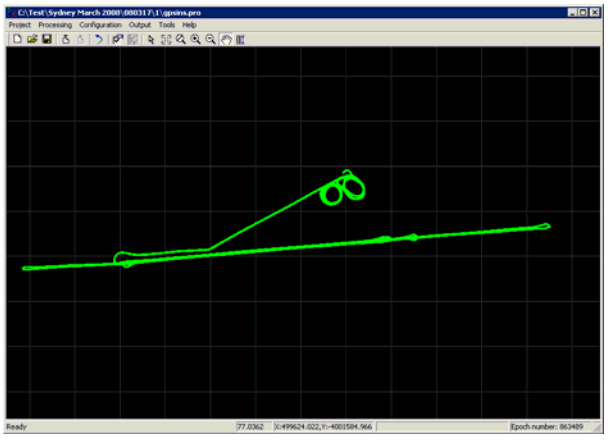


Fig. 5. The trajectory of the vehicle in the NTF test

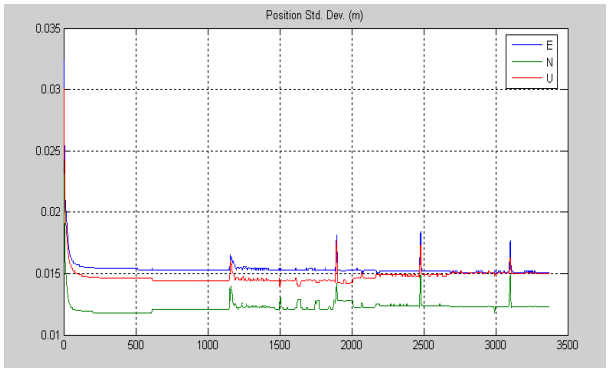


Fig. 6. The standard deviation of position in the test

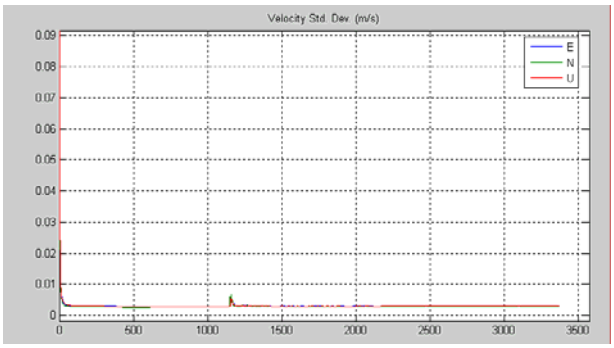


Fig. 7. The standard deviation of velocity in the test

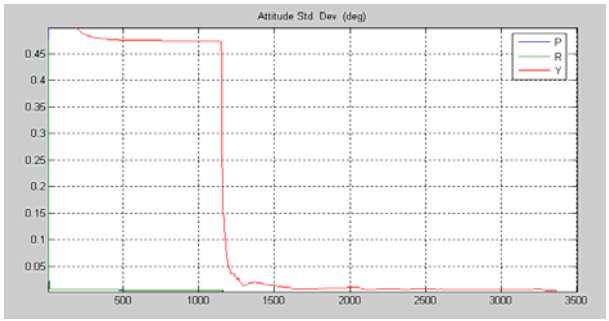


Fig. 8. The standard deviation of attitude in the test

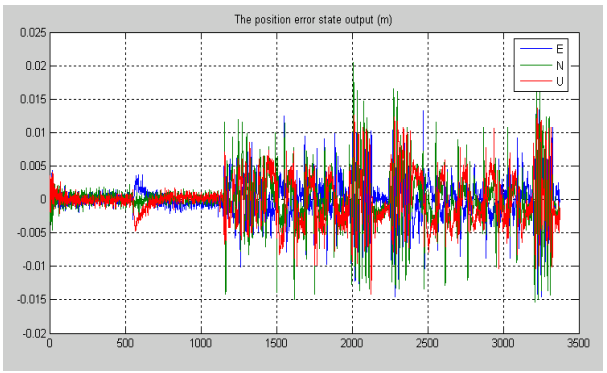


Fig. 9. The position error sub-state in the Kalman filter

In Figs. 6-8 it can be seen that the standard deviations of position and velocity are less than 0.02m and 0.01m/s respectively. The standard deviations of pitch and roll angles are less than 0.001deg as well as that of yaw, which is less than 0.01deg after the vehicle starts to move, at about the 1500th second. From Fig. 9, it can be seen that the position error sub-state output in the Kalman filter is less than 0.005m in the static part of the test, and less than 0.02m in the kinematic portion.

The Locata data was post-processed using Locata's Integrated Navigation Engine (LINE). This is an unsmoothed single point position using carrier phase measurements. The initial ambiguity bias was resolved using the data from the GPS carrier phase position. Following this initial initialization the Locata solution was computed independently of GPS. A 15m tower LocataLite location in the vicinity of the start and end of the test (indicated by the figure "eight" in Figs. 10 & 11) allowed sufficient geometry for 3D positioning using Locata. For the rest of the data where there was insufficient vertical geometry, GPS height aiding was used. Figs 10 and 11 show the independent Locata and GPS solutions (without lever arm correction) for the section of the trajectory in the vicinity, and the start and end of the test, respectively. The Locata solution compared to the GPS solution to within a few cm for the entire trajectory.

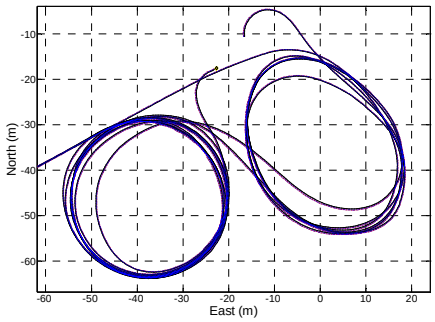


Fig. 10. Section of trajectory showing independent Locata solution (black) vs GPS (blue)

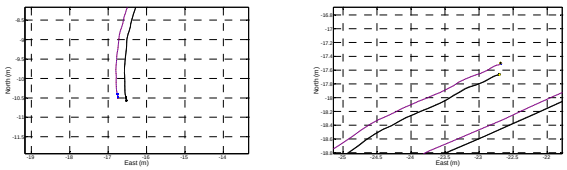


Fig. 11. Start and end of trajectory showing independent Locata solution (black) vs GPS (blue)

To test the GPS/INS/Locata integration, some GPS observation epochs were deleted to simulate two GPS blockages from 94100 to 94250 (week) seconds and from 94500 to 94600 (week) seconds. The INS standalone navigation errors with this deleted GPS data were about 40 metres and 13 metres, respectively, as shown in Fig. 11.

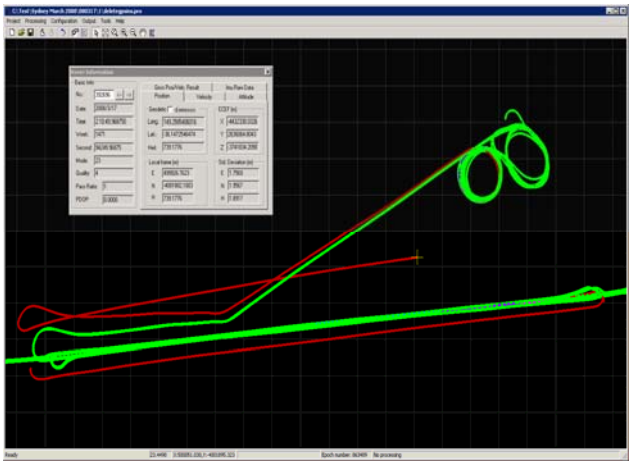


Fig. 12. The main trajectory of GPS/INS integration with deleted GPS data

In the final GPS/INS/Locata integration test, Locata compensated for the missing GPS blocks. The integration result was almost identical to the GPS/INS integration result obtained with the original GPS observed data. Fig. 12 shows the predicted residuals of the Kalman filter for the Locata measurements during the GPS outages. Fig. 13 shows the corresponding estimated residuals. In Figs. 13 and 14, it can be seen that the INS standalone navigation errors were of the order of a few centimetres.

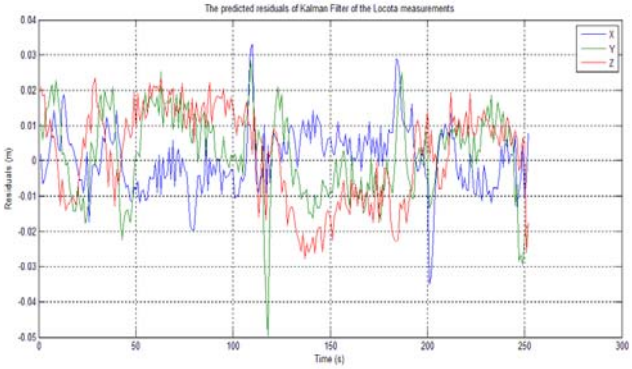


Fig. 13. The predicted residuals of the Kalman filter for Locata measurements during GPS outages

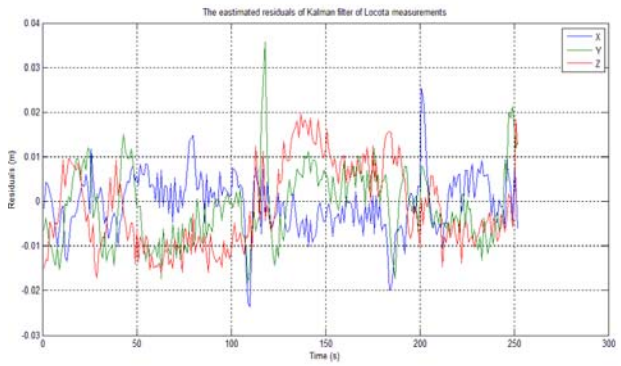


Fig. 14. The estimated residuals of the Kalman filter for Locata measurements during GPS outages

Test 2 - electric car

Early in 2007, UNSW researchers established a permanent LocataNet on the university campus. The purpose of this network was to provide a research and test facility at UNSW devoted to the Locata technology. The LocataNet setup at UNSW is illustrated in Fig. 15. It consists of four dual-frequency LocataLites situated on tops of four buildings surrounding a lawn test area. The Master LocataLite is located on the Civil Engineering building and the other three LocataLites are synchronized to it.

Currently, to be able to obtain a carrier phase position solution with Locata, the initial ambiguities need to be resolved by initializing the rover receiver on a known position. For this purpose, a point in the middle of the test area was surveyed and the coordinates were used to initialize the Locata receiver.

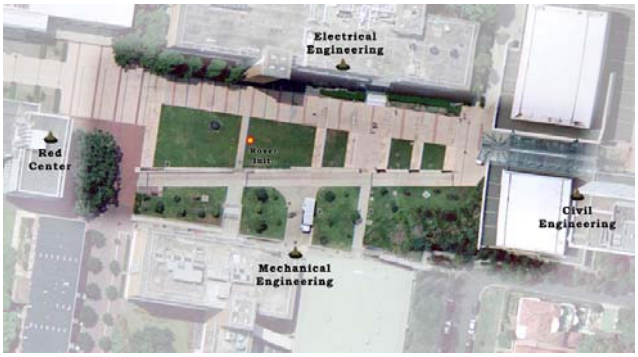


Fig. 15. LocataLites on the UNSW campus

SNAP Lab has developed a small electric car that can be driven using an attached hand-held controller. The controller enables the car to move in both forward and reverse directions, and to steer the front wheels.

For these tests the same mounting platform as the one used in the previous experiment allowed all the sensors and ancilliary equipment to be attached to the car. For this experiment the following equipment was used:

- A Locata receiver
- Omnistar GPS receiver
- Leica MC500 GPS receiver
- Boeing C-MIGIT INS
- Leica GRZ4 360 degree prism (to be tracked by a Leica Robotic Total Station)
- Two time-sync FPGA data logging devices

The starting position was the known point in the middle of the Locata network. The car was then driven in a circular path three times before finishing back at the starting position.



Fig. 16. The electronic car used in the UNSW campus test

During the test the raw data stream from the Locata receiver, the GPS receivers and the C-MIGIT INS were recorded using the time-sync data logging devices. In addition, a Robotic Total Station (RTS), which was set up at the edge of the test area, automatically tracked the prism position (the data was recorded internally).

The Locata data was post-processed using LINE to give a single point unsmoothed carrier phase solution. The initial ambiguity bias was resolved using the data from the GPS carrier phase position. Following this initial initialization the Locata solution was computed independently of GPS. Where there was insufficient vertical geometry (at the very west end of the trajectory shown in Fig. 17) GPS height aiding was used. The Locata-only solution and the RTS result are shown in Fig. 17. The two solutions compare to within a few cm of each other.

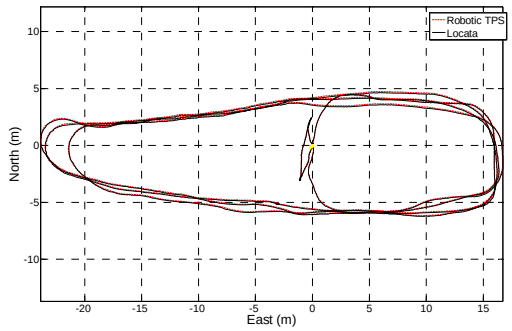


Fig. 17. The trajectory from the Locata-only and Robotic Total Station solutions

The integrated GPS/INS processing was then done. The predicted residuals and estimated residuals of the integrated solution are shown in Figs. 18 and 19.

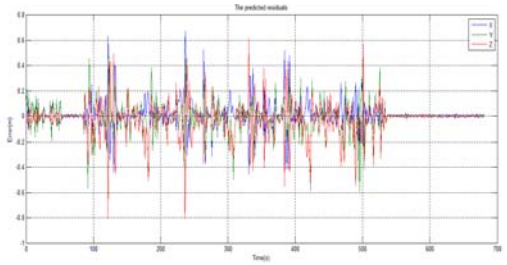


Fig. 18. The predicted residuals of the GPS/INS integration Kalman filter

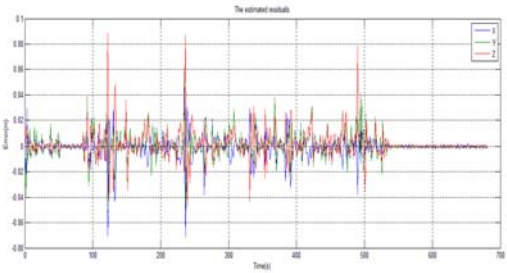


Fig. 19. The estimated residuals of the GPS/INS integration Kalman filter

In order to test the GPS/INS/Locata integration, two GPS outages were simulated by simply removing the data from the GPS file, as described in Table 2.

Table 2. Simulated GPS outages

GPS outage	Period (Week Second)
1	103703.0 – 103713.0
2	103834.0 – 103844.0

In comparison with the original GPS/INS integration, the standalone INS solution has errors of about 35m and 12m during the first and second outages, respectively.

The GPS/INS/Locata integration significantly reduced the navigation error during the GPS outages, as summarized in Table 3.

Table 3. The difference between GPS/INS/Locata (with deleted GPS data) and original GPS/INS integration solution.

GPS outages.	Parameter	Axis	Error of GPS/INS/Locata	Error of GPS/INS
1	Position	East	0.093	-19.027
		North	1.140	-26.315
		Up	0.001	-1.142
	angles	Pitch	-0.0005	-0.0014
		Roll	0.0001	0.0004
		Yaw	0.0025	-0.0079
2	Position	East	0.000	-1.257
		North	0.053	-12.315
		Up	0.001	0.001
	angles	Pitch	-0.001	0.000
		Roll	0.0007	0.0012
		Yaw	-0.021	-0.0068

From Table 3 it can be seen that 3D position differences between the GPS/INS/Locata (with deleted GPS data) and the original GPS/INS integration result have been reduced to 1.143m and 0.053m during the two GPS outages, respectively. However, the improvement in the accuracy of the attitude angles is not obvious because a 10 second GPS outage is not long enough to cause a significant INS drift.

CONCLUDING REMARKS

The test experiments described here are a demonstration of the proof-of-concept of a triple-integration GPS/INS/Locata system. The navigation results indicate that this sensor combination may support navigation in GPS-denied environments, as long as some partial view of the LocataLites within the network is available. Further development of this triple integration system is being undertaken.

ACKNOWLEDGMENTS

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